TA-CSA: A Spatiotemporal Framework With Temporal Aggregation and Clustered Spatial Attention for Traffic Forecasting

Peng Wang, Wenhao Zhang, Longxi Feng, and Kanghua Hui

***Abstract*—Accurate traffic flow forecasting is crucial in intelligent transportation systems (ITSs) for mitigating congestion and enhancing intelligent traffic management. Although recent methods have shown excellent performance, they still encounter two major limitations: 1) existing models face challenges in balancing prediction accuracy and computational efficiency and 2) existing spatial modeling methods use unified mechanisms to model all nodes, neglecting the inherent heterogeneity among them. To address these limitations, we propose a spatiotemporal architecture, TA-CSA for traffic prediction, which consists of temporal aggregation pooling (TAP) and clustered spatial attention (CSA). Specifically, in the temporal dimension, we design the TAP module. This module extracts temporal features from multiple time windows through parallel pooling operations, thereby strengthening the model's capacity to capture global temporal trends and dynamic fluctuations. In the spatial dimension, we introduce CSA. The method employs hierarchical clustering to divide nodes with similar functional characteristics into the same cluster and performs local attention calculations to achieve regional-level spatial perception. Extensive experiments on six real-world traffic datasets demonstrate that TA-CSA consistently outperforms state-of-the-art baselines in both routine and long-horizon forecasting tasks.**

***Index Terms*—Attention mechanism, hierarchical clustering, node heterogeneity, traffic flow forecasting.**

I. Introduction

WITH the rapid development of urban transportation systems, the continuous growth of vehicles has triggered numerous challenges, including increasing traffic congestion, environmental pollution, and frequent traffic accidents. To address these challenges, intelligent transportation systems (ITSs) have emerged, aiming to improve traffic management efficiency and travel safety through advanced technologies and data-driven methodologies [1]. On the one hand, ITS enhances traffic efficiency, saves a significant amount of time, and reduces fuel consumption caused by traffic congestion, thereby reducing both transportation and travel costs for residents. On the other hand, ITSs can offer the public more reliable and convenient travel services, reduce the likelihood of traffic accidents, and thereby minimize casualties. In addition, ITS can improve urban air quality, promote sustainable development, and enhance the quality of life and well-being of residents.

Previous studies typically employed statistical methods, such as the historical average (HA) [2] and the autoregressive integrated moving average (ARIMA) [3], for traffic prediction. However, the inherent complexity, nonlinearity, and spatiotemporal dependencies of traffic data hinder them from effectively capturing traffic patterns.

In recent years, deep learning methods have become a crucial technical tool in traffic prediction. In the temporal dimension, recurrent neural networks (RNNs) [4] and their variants, such as long short-term memory (LSTM) networks [5] and gated recurrent units (GRUs) [6], have been widely used for dynamic prediction of traffic due to their advantages in modeling time series data. In terms of spatial dimensions, early studies mainly used convolutional neural networks (CNNs) [7] to extract spatial correlations in traffic networks. Specifically, these methods treat the road network as a regular grid structure and represent the traffic status of each road segment as nodes in the grid, thereby capturing the spatial dependencies between different road segments through convolution operations. However, real-world road networks have a non-Euclidean structure where each sensor node may have a different number of adjacent nodes, which makes it difficult for CNN-based methods to effectively extract spatial features. To address this problem, researchers have started to employ spatiotemporal graph neural networks (STGNNs) to capture complex spatiotemporal correlations [8], [9], [10]. Different from CNNs, STGNNs are able to model complex relationships between nodes more effectively and therefore perform better in traffic prediction. In subsequent studies, researchers combine STGNN with attention to further improve performance [11], [12], [13]. Although these methods have made significant progress in capturing spatiotemporal features in traffic prediction, they still have certain limitations.

First, in the temporal dimension, most existing models struggle to maintain the balance between performance and efficiency when modeling long-term temporal dependencies. Many existing approaches utilize RNNs or 1-D convolutional neural networks (1-D CNNs), which have inherent limitations in modeling long-term temporal dependencies. In particular, RNNs often encounter problems, such as vanishing or exploding gradients, which greatly reduce their ability to model long-term dependencies in time series [14]. CNN-based methods perform forward computation by sliding convolutional kernels over 1-D continuous symbol sequences. However, their limited receptive field restricts the model's ability to capture long-range dependencies, thereby leading to performance reduction in long-time prediction tasks. To

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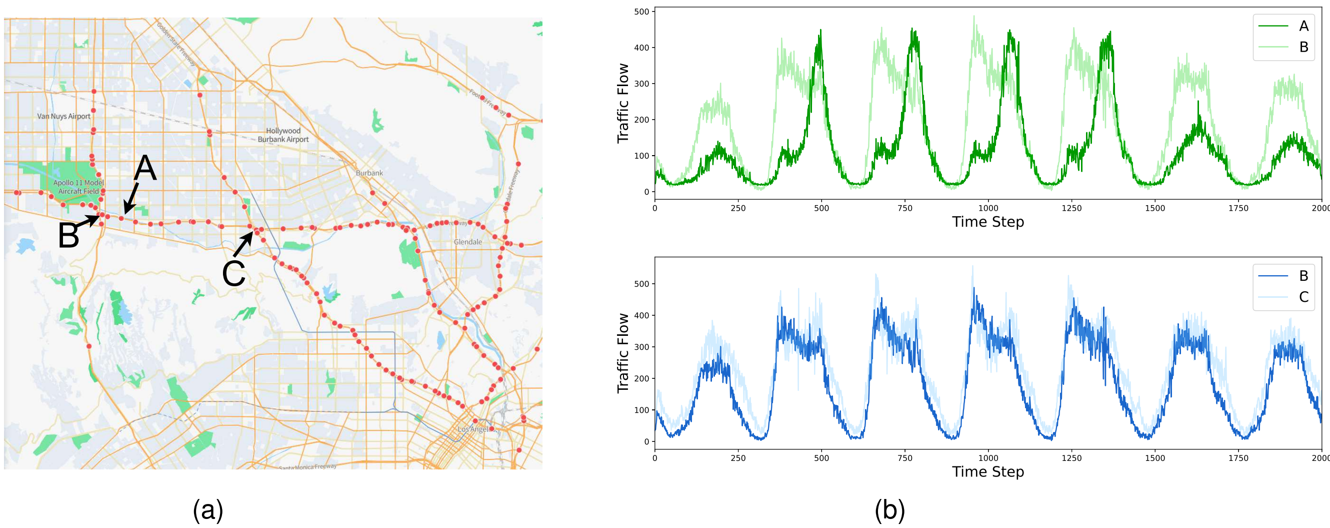

Fig. 1. (a) Physical connectivity of sensor nodes in a transportation network in geographic space. (b) Traffic flow at a node on a particular day, reflecting the distribution of traffic at different nodes.

address this issue, some researchers have employed temporal convolutional networks (TCNs) to capture long-term temporal dependencies [9], [10], [13], [14], [15], [16], [17]. Although TCNs can increase the receptive field by combining expansion factors, the layer-by-layer stacking structure may impair their ability to learn traffic patterns effectively. Moreover, to capture temporal dependencies over an extended time horizon, it must stack a sufficient number of layers, thereby reducing computational efficiency. Recent studies have explored the use of attention mechanisms in traffic prediction tasks [13], [18], [19], [20], [21]. Attention provides a flexible and efficient approach that focuses on the most relevant time steps and captures long-term dependencies without stacking multiple layers. However, despite these advantages, the computational complexity increases significantly due to the quadratic growth of pairwise computations between time steps, leading to reduced performance and efficiency when the prediction range is extended.

Second, in the spatial dimension, existing methods generally utilize uniform mechanisms to model all nodes, ignoring node heterogeneity. Spatial heterogeneity refers to the differences in structure, function, or traffic patterns of different regions. In traffic forecasting, different types of regions exhibit distinct traffic flow patterns. Most existing spatiotemporal traffic prediction models, such as STGCN [22], ASTGCN [23], and GMAN [24], typically assume that all nodes in the graph share the same structural characteristics and behavioral patterns during computation. However, they ignore the differences in functions and connectivity modes between different nodes. This limits the model's ability to express the differences among nodes in complex transportation networks, making it difficult to adequately capture the dynamic changes within the region, thus affecting the model's prediction accuracy and generalization performance.

Existing methods generally assume that physically adjacent nodes have stronger correlations, and therefore ignore the potential dependencies between geographically distant but functionally similar nodes. However, in urban road networks, sensors deployed in different functional areas, such as highways, arterials, or side roads, often exhibit significantly different traffic patterns and spatiotemporal behaviors, which results in significant spatial heterogeneity. Neglecting such structural differences often leads to generalization of the attention distribution, making it difficult for the model to capture spatial dependencies [25]. As shown in Fig. 1(a), B is topologically adjacent to A but not to C in the traffic network. However, in Fig. 1(b), the traffic flow between A and B differs significantly, while B and C are highly similar. As shown in Fig. 1, A and B are topologically adjacent to each other, but their functional attributes are significantly different. On the contrary, B and C exhibit high functional similarity even though they are not adjacent to each other in the graph structure. Meanwhile, traffic data often suffers from problems, such as sensor failures, missing data, and noise interference, which impose higher requirements for robustness on graph-based modeling methods. In global attention, abnormal information from any node may spread throughout the entire graph structure. This process amplifies noise, ultimately leading to a decline in the model's prediction performance [26]. Different from the above methods, hierarchical clustering divides nodes into areas based on Euclidean distance, which can group geographically distant but globally semantically similar nodes into the same area. Meanwhile, it restricts the attention computation to the internal area, which can effectively reduce the interference of irrelevant node characteristics and thus improve the robustness of the model in complex traffic environments.

Motivated by the above analysis, this article proposes an efficient spatiotemporal model for traffic prediction. In the temporal dimension, we design a temporal aggregation pooling (TAP) module to extract temporal features under multiple time windows through parallel pooling operations, which enhances the model's ability to capture long-term dependencies. In the spatial dimension, we introduce clustered spatial attention (CSA). The method adopts a hierarchical clustering approach to divide nodes with similar functional characteristics into multiple clusters, and performs local attention computation within the clusters, thereby achieving regional-level spatial perception. The method not only effectively utilizes the heterogeneous characteristics of the nodes but also significantly reduces the interference of anomalous nodes and noise on the model. Our main contributions can be summarized as follows.

1) We propose an efficient and robust traffic prediction model, TA-CSA, which enhances the modeling capability of long-term dependence and spatial heterogeneity through temporal aggregation and CSA.
2) We design a TAP module to aggregate temporal features across multiple time windows, which significantly enhances the model's ability to capture global trends and perceive the dynamic changes of time. In the spatial dimension, we introduce CSA. The method employs hierarchical clustering to divide nodes with similar functional characteristics into multiple clusters and performs local attention computations within clusters to achieve regional-level spatial perception. The method effectively utilizes the node heterogeneity and improves the robustness of the model.
3) We have conducted extensive experiments on six real-world datasets, and the results show that our proposed TA-CSA achieves state-of-the-art performance.

The structure of this article is as follows. Section II reviews related work on traffic prediction. Section III defines the problem. Section IV details the method of our model. Section V presents experiments conducted on six real-world datasets and analyzes the results. Finally, Section VI concludes the article and discusses future research directions.

II. Related Work

A. Traditional Methods for Traffic Forecasting

Early studies treated traffic prediction as a time series problem and employed classical modeling techniques, such as HA, support vector regression (SVR) [27], and Kalman filter (KF) [28]. However, many of these traditional methods (e.g., ARIMA and KF) are based on linear assumptions, thereby limiting their ability to capture complex nonlinear patterns common in traffic data. Moreover, traffic systems naturally exhibit spatial structures characterized by strong interdependencies among different locations. Traditional approaches tend to emphasize temporal patterns while neglecting these spatial correlations, which often results in suboptimal prediction performance.

B. Spatial–Temporal Neural Networks for Traffic Forecasting

With the increase in large-scale traffic data and the advancement of computational ability, deep learning methods have gained significant attention in the field of traffic forecasting. These advances enable models to handle massive datasets, utilize more complex neural structures, and achieve better computational efficiency. Ma et al. [29] were the first to apply the LSTM network to the traffic prediction task. Similarly, Duan et al. [30] employed LSTM models for travel time estimation, further demonstrating the potential of RNNs in capturing temporal dependencies in traffic data. To further enhance the modeling of spatiotemporal correlations in traffic data, researchers have explored hybrid deep learning architectures. Zhang et al. [31] introduced ST-ResNet, a deep CNN designed to capture both spatial and temporal correlations. Liu et al. [5] proposed conv-LSTM, which utilizes LSTM and CNN for extracting spatiotemporal features. However, CNN-based models are designed for regular grid inputs (e.g., images) and exhibit inherent limitations when applied to non-Euclidean structured traffic data. To address this issue, subsequent studies employed graph neural networks (GNNs) to more effectively capture spatial dependencies in non-Euclidean structures. Li et al. [8] proposed diffusion convolutional RNNs (DCRNNs), which model spatial correlations through bidirectional random walks on graph structures. Guo et al. [32] developed GC-LSTM, a sequence-to-sequence framework that integrates spectrogram convolution with LSTM to jointly learn spatial and temporal features for traffic speed prediction.

To address the issues of gradient vanishing and explosion in RNN-based models, researchers introduced TCNs to capture temporal dependencies. Yu et al. [33] combined TCNs with GCNs to effectively extract spatiotemporal features. Wu et al. [9] proposed a self-learning adjacency matrix construction method based on node embeddings, and combined GCN with dilated causal convolution to improve the performance of traffic prediction. Zhao et al. [34] proposed T-GCN, a hybrid framework that combines GCNs with gated recurrent units (GRUs). Since many existing models rely on predefined or static adjacency matrices, they often struggle to capture the complex and evolving spatial dependencies in real-world traffic systems. To address this limitation, recent studies have shifted toward learning data-driven graph structures that can dynamically adapt to changing traffic patterns. For instance, Li and Zhu [16] constructed dynamic graphs based on temporal similarity between nodes, enabling the model to reflect time-varying spatial correlations. Han et al. [10] assumed that shared graphs can represent recurring traffic patterns at the same time across different days. They introduced three learnable matrices along with a core tensor to model time-specific spatial dependencies among road segments. These components allow the model to perform message passing over the dynamic adjacency matrix, thereby aggregating hidden states from adjacent nodes in a temporally adaptive manner. Ta et al. [26] proposed Ada-STNet to learn the graph structure at both micro and macro levels to capture comprehensive spatial correlations. Guo et al. [18] computed the strength of dynamic spatial dependencies among nodes and employed an element-wise dot product to adjust the static weight matrices to capture dynamic spatial relationships. Lan et al. [35] computed the spatial and temporal distances from traffic data by constructing a data-driven spatiotemporal perception dynamic graph. However, convolution-based models (e.g., CNNs and GCNs) depend on a large number of matrix operations and parameter updates during the training process, leading to significantly higher computational resource consumption while reducing computational efficiency.

With the advancement of attention mechanisms, it has been successfully extended beyond natural language processing to various other domains, notably traffic forecasting. In this field, attention has been proven to be remarkably effective in capturing complex temporal and spatial dependencies by allowing models to prioritize key information patterns in traffic data. Liu et al. [36] designed the STAEformer model, which utilizes a novel adaptive embedding and spatiotemporal attention to effectively capture traffic patterns. Jiang et al. [37] considered temporal propagation delays and designed an attention-based spatial block to capture spatial dependencies. In addition, Sun et al. [38] proposed a novel decoupled spatiotemporal embedding structure, which effectively integrates

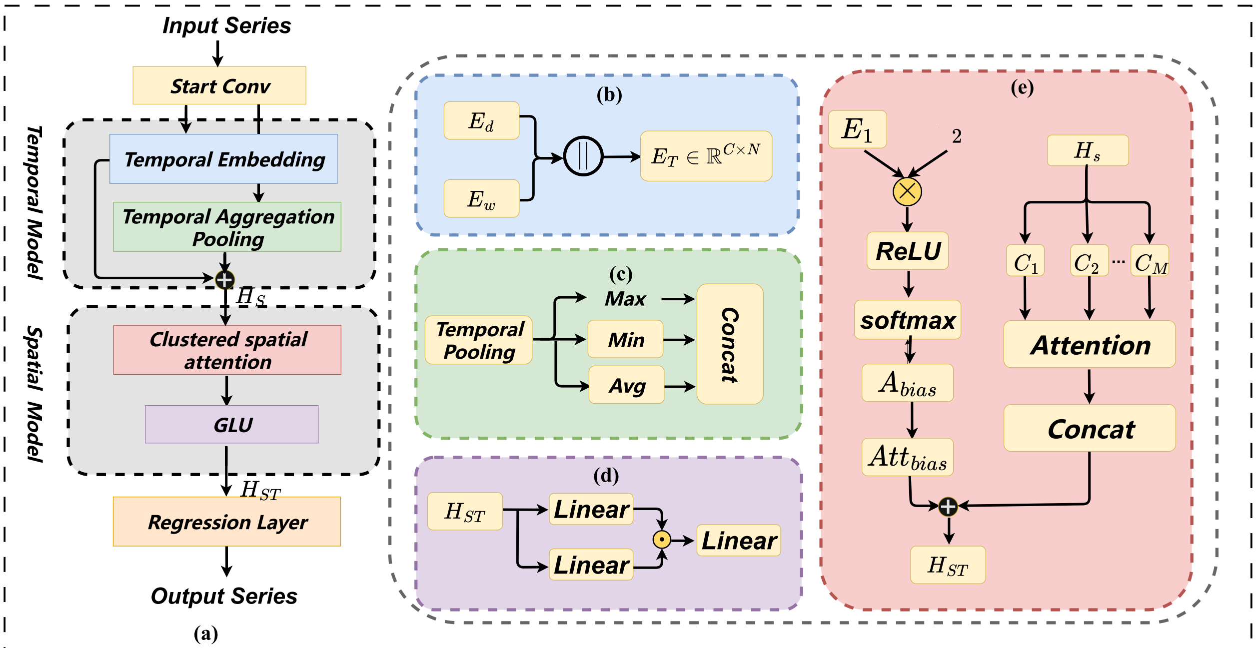

Fig. 2. (a) Framework of TA-CSA. (b) TE. It contains two parts: the days-of-the-week embedding matrix (E_w) and the moments-of-the-day embedding matrix (E_d). (c) TAP. Extract time features using the maximum, minimum, and average aggregation operations. (d) GLU. Filter key features, balance linear and nonlinear transformations. (e) CSA. It employs hierarchical clustering to divide nodes with similar functional characteristics into multiple clusters and performs local attention computations within these clusters.

low-frequency features into sequences and enhances the ability of the model to capture long-term dependencies. Cai et al. [39] employed dual-graph masked spatial attention to model complex spatial dependencies and utilized trend-aware multihead attention to extract temporal features. However, attention-based approaches usually employ a uniform mechanism to model all nodes, ignoring the differences in functions and connectivity of different nodes, thus affecting the prediction accuracy and generalization performance of the model.

Compared with the above studies, the TA-CSA proposed in this article has the following advantages and differences.

1) TA-CSA is designed with a TAP module that significantly enhances the model's ability to capture global trends and perceive the dynamic changes of time, achieving more efficient long-term forecasting performance.
2) TA-CSA employs a CSA in the spatial dimension. It divides nodes with similar functional characteristics into multiple clusters and performs local attention computation within clusters to achieve regional-level spatial perception.

III. Problem Statement

A. Notations and Definitions

1) Traffic Network: A road network can be represented as a graph $G = (V, E, A)$, where V is the set of nodes, with $|V| = N$ denoting the number of nodes, E is the set of edges, and $A \in \mathbb{R}^{N\times N}$ represents the connectivity between nodes.

2) Traffic Signals: The historical traffic series at time slice t is represented as a graph signal $X_t \in \mathbb{R}^{N\times F}$, where F denotes the dimension of each node's attributes, such as traffic speed, volume, or occupancy.

B. Problem Definition

1) Traffic Forecasting: Given T historical graph signals $X = [X_{(t-T+1)}, \ldots, X_{(T)}] \in \mathbb{R}^{T\times N\times F}$, our goal is to learn a function F to forecast T future graph signals $Y = [Y_{(t+1)}, \ldots, Y_{(t+T)}] \in \mathbb{R}^{T\times N\times F}$ as accurately as possible. The forecasting process can be formulated as follows:

$$\left[X_{(t-T+1)}, \ldots, X_{(T)}\right] \xrightarrow{F(\cdot)} \left[Y_{(t+1)}, \ldots, Y_{(t+T)}\right]. \tag{1}$$

IV. Methodology

In this section, we first introduce the architecture of the proposed TA-CSA, as shown in Fig. 2. Our TA-CSA consists of three main modules: temporal embedding layer (TE), TAP, and CSA.

The TE module improves the model's ability to capture periodic and trend patterns of traffic data through adaptive learnable embedding. The TAP module aims to capture temporal features over multiple time windows. The model captures both general trends and abnormal fluctuations more effectively by aggregating temporal features across multiple time windows. The CSA module employs hierarchical clustering to divide nodes with similar functional characteristics into multiple clusters and performs local attention computations within these clusters to achieve area-level spatial perception. Details of each component are shown below.

A. Temporal Embedding Layer

Traffic time-series data typically show a periodic pattern of evolution, with morning and evening peaks during the day and a relatively flat pattern at other times of the day. In order to better capture and memorize the traffic patterns during different periods, we introduce a time-embedding mechanism to highlight the effects of periodicity and trends on traffic flow.

As shown in Fig. 2, the embedding layer consists of two sets of learnable dictionary matrices, namely, the days-of-the-week embedding matrix $E_w \in \mathbb{R}^{7\times\varphi}$ and the moments-of-the-day embedding matrix $E_d \in \mathbb{R}^{t_w\times\varphi}$, where seven denotes the number of days in a week, t_w denotes the number of time steps in a day, and φ is the embedding dimension.

We denote W_t, $D_t \in \mathbb{R}^T$ as the day-of-week and time-of-day data for the traffic series from time step $(t - T + 1)$ to t. Through them, day-of-week embedding $E_w^t \in \mathbb{R}^{T\times\varphi}$ and the time-of-day embedding $E_d^t \in \mathbb{R}^{T\times\varphi}$ are obtained from the corresponding learnable embedding dictionaries. Finally, they broadcast in the embedding dimension and spliced along the channel dimension by the following equation:

$$E_{T[,\ldots,-1]} = \text{Concat}\left(E_d^t, E_W^t\right) \tag{2}$$

where $E_{T[,\ldots,-1]} \in \mathbb{R}^{C\times N}$ is final temporal embedding, and $T[,\ldots,-1]$ denotes the last time step of the encoded sequence in the temporal context, which is employed as a unique time identifier for each traffic signal within the current time window.

B. Temporal Aggregation Pooling

Pooling-based time window feature aggregation methods have advantages over convolution or attention mechanisms in terms of computational efficiency, simplicity of implementation, and parameter independence. By directly extracting statistical features (such as mean and extreme values) within the window, they effectively suppress noise and reduce computational overhead. Therefore, we employ the pooling operation as a means of temporal feature aggregation and design the TAP module.

Specifically, we divide the original time series into multiple nonoverlapping time windows and perform multiple pooling operations within each window to extract temporal features. This module performs parallel average, maximum, and minimum pooling operations, concatenates the pooling results within each time window, and then completes the feature fusion and dimensionality reduction using 1-D convolution. Average pooling, maximum pooling, and minimum pooling are used to capture the overall trend of traffic flow, peaks in flow (e.g., extreme congestion during morning and evening peaks), and troughs in flow (e.g., low-flow periods at night), respectively. Multiscale temporal features are aggregated by stacking pooling layers with different window sizes, which enables the model to capture traffic features at multiple temporal scales and enhances its ability to model long-term dependencies.

Mathematically, given the original input $X \in \mathbb{R}^{F\times N\times T}$, where F denotes the feature dimension, N is the number of nodes, and T is the time step. After the start convolution layer, we obtain the high-dimensional representation $H \in \mathbb{R}^{C\times N\times T}$. The formula in a specific temporal block can be expressed as follows:

$$\begin{aligned} X_{\text{avg}} &= \text{AvgPool}(X) \\ X_{\max} &= \text{MaxPool}(X) \\ X_{\min} &= \text{MinPool}(X) \end{aligned} \tag{3}$$

where X_{avg}, $X_{\max}$, and $X_{\text{nin}} \in \mathbb{R}^{C\times N\times T'}$denote the output tensor after average pooling, maximum pooling, and minimum pooling, respectively.$T' = (T/K)$ denotes the time step after pooling, and K is the size of the time window.

Next, the three pooling results are concatenated along the time dimension, and then the multiscale features are adaptively fused by 1-D convolution to obtain $H_t \in \mathbb{R}^{C\times N}$

$$H_t = \text{con}\nu\left(\text{Concat}\left(X_{\text{avg}}, X_{\max}, X_{\min}\right)\right). \tag{4}$$

Finally, we connect H_t with the temporal embedding E_T to obtain the updated spatiotemporal representation $H_s \in \mathbb{R}^{2C\times N}$

$$H_s = \text{Concat}(E_T, H_t). \tag{5}$$

C. Clustered Spatial Attention

In traffic prediction, existing spatial modeling methods often employ uniform mechanisms for all nodes, overlooking node functional differences, and are prone to anomalies or interference from missing data, resulting in unstable performance. To address this problem, we propose a CSA modeling method. The method uses hierarchical clustering to divide nodes with similar functional characteristics into multiple clusters and performs local attention computation within these clusters to achieve regional-level spatial perception as well as to restrict the spread of abnormal information.

Hierarchical clustering compresses the high-dimensional node feature representation of the data into a low-dimensional feature vector. The vectors are averaged over the batch dimensions to generate a stable global representation for each node, which improves the stability and consistency of clustering and prevents the fluctuation of individual samples from affecting the clustering results of the attention. The calculation formula is as follows:

$$f = \frac{1}{B}\sum_{b=1}^{B} H_s[b,:,:,:] \tag{6}$$

where $f \in \mathbb{R}^{N\times 2C}$ denotes the feature matrix.

Then, the nodes are clustered by hierarchical clustering methods. The specific steps are as follows: first, initialize the clustering structure, and treat each node individually as a cluster, with a total of N clusters, $\{C_1, C_2, \ldots C_M\}$, where $C_i = \{x_i\}$ denotes a cluster, and x_i is the feature vector of the i node. The distances between all clusters are computed to generate a distance matrix $d \in \mathbb{R}^{N\times N}$. Select the two clusters with the smallest distance and merge them into a new cluster. Then, calculate the distance between this new cluster and the other clusters, and update the distance matrix. Repeat this process until a predefined number of clusters is formed. The formula is as follows:

$$d_{ij} = \|x_i - x_j\|_2 \tag{7}$$

where d_{ij} is the similarity between nodes i and j.

After completing the merging of clusters, attention computation is performed within each cluster, and finally, the attention of all clusters is merged. The formula is as follows:

$$\begin{aligned} Q_c, K_c, V_C &= W_Q H_s, \quad W_K H_s, \quad W_V H_s \\ \text{Attention}_{\text{clu}} &= \text{softmax}\left(\frac{Q_C K_C^T}{\sqrt{d_k}}\right) V_c. \end{aligned} \tag{8}$$

Finally, we introduce an adaptive adjacency matrix to provide a learning bias for the attention mechanism, which improves the structural representation of the attention mechanism when dealing with situations, such as long-distance dependence and weakly connected nodes. The formula is expressed as

$$A_{\text{apt}} = \text{softmax}\left(\text{ReLU}\left(E_1 \cdot E_2^T\right)\right)$$
$$\text{Att}_{\text{bias}} = A_{\text{apt}} \cdot V_{\text{bias}} \tag{9}$$

where E_1 and $E_2 \in \mathbb{R}^{N\times\varphi}$ are learnable node embeddings, N is the number of nodes, φ is the hidden dimension, A_{apt} is the adaptive adjacency matrix, ReLU and softmax denote the activation function, and Att_{bias} is a learnable structural bias.

Combining CSA with learnable structural biases. The formula is shown below

$$H_{\text{ST}} = \text{Attention}_{\text{clu}} + \text{Att}_{\text{bias}}. \tag{10}$$

D. GLU and Prediction Layer

We introduce gated linear units (GLUs) for filtering key features, balancing linear and nonlinear transformations, and fusing multiple pieces of information through gating mechanisms. The formula is shown below

$$H_Y = \text{GLU}\left(H_{\text{ST}}\right) W_m \tag{11}$$

where H_{ST} is the output of spatial attention, and $W_m \in \mathbb{R}^{2C\times 2C}$ is the parameter matrix of the learnable linear transformation. Finally, the results are predicted by MLP

$$Y = \text{MLP}\left(H_Y\right) \tag{12}$$

where $Y \in \mathbb{R}^{F\times N\times T}$ is the result of the traffic forecast.

V. Experiments

In this section, we conduct experimentation across six real-world datasets to answer six research inquiries.

RQ1: How does the performance of TA-CSA compare with current state-of-the-art models?

RQ2: How does TA-CSA perform in terms of efficiency compared to the baseline model?

RQ3: How do different components affect the performance of TA-CSA?

RQ4: How do the pooling methods in TAP work in the model?

RQ5: How does the number of clusters affect the performance of TA-CSA?

RQ6: How robustly does the TA-CSA model perform under different experimental conditions?

A. Datasets

To verify the effectiveness of the proposed method, we conducted experimental analysis on six real-world traffic datasets. Among them, the PEMS03, PEMS04, PEMS07, and PEMS08 datasets represent coarse-grained highway traffic scenarios [39]. In contrast, the NYCTaxi and NYCBike datasets correspond to fine-grained urban traffic demand scenarios [40], which allow a comprehensive evaluation of the model under diverse traffic conditions.

PEMS03, PEMS04, PEMS07, and PEMS08 are derived from the performance measurement system (PEMS) of the California Department of Transportation (Caltrans) and cover three key metrics: traffic speed, traffic volume, and road occupancy. In contrast, NYCBike and NYCTaxi focus on intracity transportation demand data; NYCBike reflects the short-distance travel patterns of city residents by recording the number of bikeshare bikes placed and returned at each station; NYCTaxi data contains detailed trip information for cab services within New York City, including passengers' pick-up and drop-off locations. These datasets offer a multilevel view, ranging from highways to city streets, and from traffic states to travel behaviors, which enables a comprehensive evaluation of the model's adaptability and robustness in various transportation environments. Further statistical information is detailed in Table I.

TABLE I
Detailed Statistics of Six Datasets

Dataset	Nodes	Interval	Samples	Time Range
PEMS03	358	5 min	26208	09/01/2018–11/30/2018
PEMS04	307	5 min	16992	01/01/2018–02/28/2018
PEMS07	883	5 min	28224	05/01/2017–08/31/2017
PEMS08	170	5 min	17856	07/01/2016–08/31/2016
NYCBike	250	30 min	4368	04/01/2016–06/30/2016
NYCTaxi	266	30 min	4368	04/01/2016–06/30/2016

B. Metrics

For the metrics, three commonly used performance metrics are employed in this work: mean absolute error (MAE), root-mean-square error (RMSE), and mean absolute percentage error (MAPE). MAE represents the average of the absolute values of the predicted and true values. RMSE represents the root-mean-square error between the predicted and true values. MAPE is the mean absolute percentage error. The metrics are defined below

$$\text{MAE} = \frac{1}{M}\sum_{i=1}^{M}|\hat{Y} - Y| \tag{B-1}$$

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum\nolimits_{i=1}^{M}\left(\hat{Y} - Y\right)^2} \tag{B-2}$$

$$\text{MAPE} = \frac{100}{M}\sum\nolimits_{i=1}^{M}\left|\frac{\hat{Y} - Y}{Y}\right| \tag{B-3}$$

where Y denotes the true value, $\hat{Y}$ denotes the predicted value, and M denotes the number of values to be predicted.

C. Baseline Models

To verify the validity of our method, we compare TA-CSA with the classical methods and state-of-the-art methods.

1) *HA [2]:* HA is a traditional time series forecasting method, and its core idea is to use the average traffic state during the same historical period as the predicted value for the corresponding future moment.
2) *VAR [41]:* VAR is a multivariate time series modeling approach that enables joint forecasting of future trends in multiple time series by modeling the linear interdependencies among the variables.
3) *STGCN [33]:* STGCN is a spatiotemporal modeling method that combines graph convolution and temporal convolution to capture spatiotemporal dependencies in traffic data using graph convolution and temporal convolution.
4) *DCRNN [8]:* DCRNN models the interaction between nodes in a transportation network as a diffusion process

and introduces a diffusion convolution operation to capture the spatial dependencies between nodes.

5) *GraphWaveNet [9]:* Graph WaveNet is an STGNN model that combines adaptive graph structure learning with an extended causal convolution mechanism to efficiently capture spatiotemporal dependencies.
6) *ASTGCN [23]:* ASTGCN leverages temporal attention and spatial convolution to capture dynamic spatial–temporal correlations.
7) *AGCRN [42]:* AGCRN utilizes an adaptive graph learning module and GRU to capture spatiotemporal dependencies.
8) *STG-NCDE [43]:* STG-NCDE is a spatiotemporal modeling method that combines GNNs and neural control differential equations, aiming to capture spatiotemporal dependencies.
9) *ASTGNN [18]:* ASTGNN uses a self-attention in local context and dynamic graph convolution to capture spatiotemporal correlations.
10) *STAMT [40]:* STAMT enhances spatiotemporal correlation modeling capabilities by introducing a memory repository mechanism for caching and retrieving historical traffic patterns. Simultaneously, it effectively reduces computational overhead through an efficient temporal aggregation module.
11) *Ada-STNet [26]:* Ada-STNet employs a graph structure learning component to obtain the optimal graph adjacency matrix from both macroscopic and microscopic perspectives and learns the spatiotemporal dependencies using a dedicated spatiotemporal convolutional architecture.
12) *DGCRN [44]:* DGCRN utilizes hypernetworks to extract dynamic features from node attributes, generate dynamic graph structures, and incorporate RNNs for temporal modeling to achieve traffic prediction.
13) *PDFormer [37]:* PDFormer introduces a delay-aware feature transformation module to model temporal delays and utilizes a graphical masking matrix to capture short- and long-term spatial dependencies.
14) *STAEFormer [36]:* STAEformer uses spatiotemporal adaptive embedding to capture the intrinsic spatiotemporal dependencies.
15) *PGCN [45]:* PGCN constructs progressive adjacent matrices and adopts expanded causal convolution to capture spatiotemporal dependencies.
16) *STPGNN [17]:* STPGNN introduces a key node identification module and a key graph convolution module to capture the spatiotemporal dependence of nodes.

D. Experimental Setups

The model was trained in a Pytorch environment with an RTX 3090 with 24 GB (GPU) and a 14 vCPU Intel[1] Xeon[1] Gold 6330 CPU @ 2.00 GHz (CPU). For the regular traffic prediction task, we set the input and output lengths to 12, and for the long-term task, the length is set to 60. Batch size is set to 64, learning rate to 0.001, weight decay to 0.0001, dropout rate to 0.1, training epochs to 500, and early stopping patience to 100. We use the Ranger21 optimizer [46] during training. The model uses two-layer time blocks with window sizes of 3 and 4, respectively, with feature dimension C set to 128. In addition, we randomly scrub the training data at the beginning of each epoch to ensure that the model learns the underlying patterns.

[1]Registered trademark.

E. Performance Comparison (RQ1)

In this section, we design two groups of experiments to validate the model for regular traffic prediction and long-term traffic prediction tasks to comprehensively evaluate the prediction performance of the model. Regular traffic prediction is usually performed on standard datasets that are designed to measure the basic performance of the model in predicting short-term traffic states. This task is particularly critical for real-time traffic applications to support the rapid response of the system to dynamic traffic changes. In contrast, long-term traffic forecasting focuses on evaluating the model's ability to consistently maintain forecast accuracy over a longer time horizon (e.g., the next 60 time steps). These predictions can serve application scenarios, such as route optimization, infrastructure development, and urban traffic management. Below are the results and analysis of the two types of experiments.

1) Results and Analysis on Regular Traffic Forecasting: Tables II and III show a comparison of the model's prediction performance on six traffic datasets. The results and analysis are as follows: deep learning models outperform traditional statistical methods (e.g., HA and VAR) on highway datasets. Earlier models, such as STGCN and DCRNN, depend on static adjacent matrices, which are difficult to adapt to dynamic spatial relationships, leading to limited performance. AGCRN introduces adaptive adjacent matrices, which improve the ability of modeling dynamic relationships among nodes, and outperforms GraphWaveNet and ASTGCN. STG-NCDE combines GNNs with controlled differential equations to accurately model continuous spatiotemporal dynamics. Ada-STNet constructs optimal adjacent matrices at macro and micro levels through the graph structure learning component and learns dependencies through spatiotemporal convolution. PDFormer uses a delay-aware feature transformation and a graph masking mechanism to improve performance. STPGNN introduces the key node identification and graph convolution modules and achieves excellent performance on the PEMS03 dataset.

On the traffic demand dataset, we only compare the performance of deep learning methods. Transformer-based models (e.g., STAEFormer and PDFormer) show excellent performance on both datasets. Their performance is largely due to the attention mechanism, which can adapt to both origin-destination trajectory data and graph-structured traffic flow data. Convolution-based models (e.g., STPGNN and PGCN) perform well on highway datasets but poorly on traffic demand datasets. This can be attributed to the fact that convolutional networks are effective at capturing stable and well-defined graph-structured spatiotemporal dependencies among sensor nodes. However, the interarea flow relationships in traffic demand datasets are more complex and lack fixed adjacencies, leading to their poor performance. Although GraphWaveNet is also a model based on convolutional structures, it can better capture the complex and variable flow patterns on demand datasets through adaptive adjacency matrices and multiscale time-series convolution.

TABLE II
PERFORMANCE COMPARISON ON HIGHWAY DATASETS (12 TIME STEPS). BOLD: BEST AND UNDERLINE: SECOND BEST

Method	PEMS03			PEMS04			PEMS07			PEMS08		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
HA	31.58	52.39	33.78%	38.03	29.24	27.88%	45.12	65.64	24.51%	34.86	59.24	27.88%
VAR	23.65	38.26	24.51%	28.70	44.56	19.20%	50.22	75.63	32.22 %	19.19	29.81	13.10%
STGCN	17.55	30.42	17.34%	21.16	34.89	13.83%	25.33	39.34	11.21%	17.50	27.09	11.29%
DCRNN	18.39	30.55	20.24%	21.22	33.44	14.17%	25.22	38.61	11.82%	16.82	26.35	10.92%
GraphWaveNet	19.12	32.75	18.86%	24.89	39.66	17.29%	26.39	41.52	11.96%	18.28	30.05	12.15%
ASTGCN	18.29	30.80	17.05%	22.93	35.22	16.56%	25.98	39.65	11.84%	18.86	28.74	11.15%
STAMT	14.61	23.96	15.33%	18.07	29.53	12.11%	19.28	32.32	8.14%	13.45	22.70	8.86%
AGCRN	15.20	26.57	14.64%	19.95	32.91	13.16%	20.57	34.40	8.74%	15.81	25.04	10.19%
STG-NCDE	15.57	27.09	15.06%	19.21	31.09	12.76%	20.53	33.84	8.80%	15.45	24.81	9.91%
ASTGNN	14.81	24.91	14.83%	18.74	30.95	12.47%	19.26	32.75	8.54%	15.32	24.61	9.64%
Ada-STNet	14.49	25.11	**14.35%**	18.65	29.95	13.09%	21.44	34.26	9.92%	14.21	23.02	9.29%
DGCRN	14.51	24.81	15.11%	19.22	30.58	13.44%	20.96	33.35	9.55%	14.97	23.72	9.78%
PDFormer	14.76	25.56	15.51%	18.25	29.88	12.44%	19.83	32.87	8.52%	13.58	23.50	9.04%
STAEFormer	15.18	25.80	15.33%	18.21	30.11	**12.00%**	**19.15**	32.61	**8.02%**	13.55	23.01	8.98%
PGCN	14.94	25.54	15.06%	18.68	30.08	13.07%	20.77	33.81	8.99%	14.22	23.00	9.30%
STPGNN	**14.41**	24.48	15.10%	18.57	29.89	12.97%	20.00	32.82	8.49%	14.17	23.25	9.21%
TA-CSA	14.48	**22.93**	15.55%	**17.91**	**29.35**	12.23%	19.25	**32.30**	8.16%	**13.42**	**22.42**	**8.84%**

TABLE III
PERFORMANCE COMPARISON ON TRAFFIC DEMAND DATASETS (12 TIME STEPS). BOLD: BEST AND UNDERLINE: SECOND BEST

Method	NYCTaxi pick			NYCTaxi drop			NYCBike pick			NYCBike drop		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
STGCN	6.11	11.04	41.03%	5.76	11.95	39.75%	2.17	3.62	55.76%	2.10	3.31	51.19%
DCRNN	5.88	10.71	37.09%	5.19	9.63	37.78%	2.09	3.30	54.22%	1.96	2.94	51.42%
GraphWaveNet	5.88	10.09	42.67%	5.25	9.29	36.46%	2.18	3.47	56.25%	2.09	3.26	53.78%
ASTGCN	6.78	14.92	47.88%	6.55	12.75	42.80%	2.14	3.97	61.73%	2.04	3.60	60.09%
AGCRN	6.21	12.55	41.50%	5.91	14.00	40.89%	2.07	3.20	**51.92%**	2.14	3.46	56.56%
STG-NCDE	6.11	11.35	47.37%	5.99	10.62	43.08%	2.28	4.15	62.59%	2.18	3.85	61.66%
ASTGNN	5.91	10.71	40.17%	6.28	12.05	49.78%	2.27	3.34	57.23%	2.36	3.67	60.09%
STAMT	4.98	8.74	**33.26%**	4.79	8.43	34.29%	1.99	3.10	52.87%	1.88	2.81	50.43%
Ada-STNet	5.55	9.70	36.95%	5.45	9.73	36.55%	2.22	3.60	53.71%	2.05	3.18	50.73%
DGCRN	5.63	9.96	39.34%	5.45	9.73	36.43%	2.52	3.99	58.76%	2.47	3.65	60.86%
PDFormer	5.47	9.33	37.83%	5.12	8.76	37.45%	2.29	3.58	59.23%	2.18	3.29	56.83%
STAEFormer	5.20	9.17	35.99%	4.95	8.81	33.98%	2.17	3.42	56.21%	2.14	3.30	54.75%
PGCN	6.49	11.54	42.95%	5.68	9.97	40.28%	2.29	3.72	56.46%	2.19	3.47	54.44%
STPGNN	7.04	13.16	53.40%	7.31	13.46	52.54%	2.28	3.70	57.35%	2.16	3.38	54.88%
TA-CSA	**4.96**	**8.59**	34.49%	**4.72**	**8.26**	**33.92%**	**1.94**	**3.01**	52.34%	**1.84**	**2.71**	**50.14%**

From the table, we can draw the following conclusions.

1) Our proposed TA-CSA model achieves optimal or near-optimal performance on multiple datasets. Its advantage lies in the introduction of the TAP module, which can effectively capture the trend and fluctuation changes of traffic sequences. Meanwhile, the CSA we designed can effectively utilize the spatial heterogeneity of nodes to achieve regional-level spatial perception and further improve the prediction accuracy of the model.
2) The traditional HA and VAR methods have significantly higher prediction errors on these six datasets, showing obvious performance disadvantages, indicating that they struggle with complex nonlinear spatiotemporal traffic prediction tasks.
3) Compared with models that rely on static graph structures (such as STGCN, DCRNN, and GraphWaveNet), models that simulate dynamic spatial structures (such as ASTGNN, AdaSNTet, and TA-CSA) can more accurately capture dynamic spatial correlations and achieve better performance.
4) Although some methods (e.g., PGCN and STPGNN) perform well on highway datasets, they perform poorly when dealing with urban traffic demand type datasets. This phenomenon indicates that the model has certain advantages in coarse-grained traffic scenarios, but still has limitations in fine-grained, highly dynamic modeling of urban traffic demand.

In Fig. 3, we visualize the prediction errors for six datasets at each time step. The results show that our model has smoother error fluctuations over the entire time series, further validating its stability and generalizability in standard traffic prediction tasks.

2) Results and Analysis on Long-Term Traffic Forecasting: To further validate the effectiveness of TA-CSA in the long-term traffic prediction task, we conducted long-term prediction experiments on the PEMS04 and PEMS08 datasets, and the specific results are shown in Table IV. In addition, to assess the model's stability overextended forecasting horizons, we calculate its long-term prediction performance decay from 12 to 36, 48, and 60 time steps.

From the table, we can observe that the overall performance of all methods decreases significantly as the prediction time horizon extends from short-term to long-term. This phenomenon can be attributed to the increased uncertainty associated with longer historical time windows, leading to error accumulation. Table V demonstrates the error increments from short-term to long-term forecasts for the three methods described above. To better observe the change of error over the

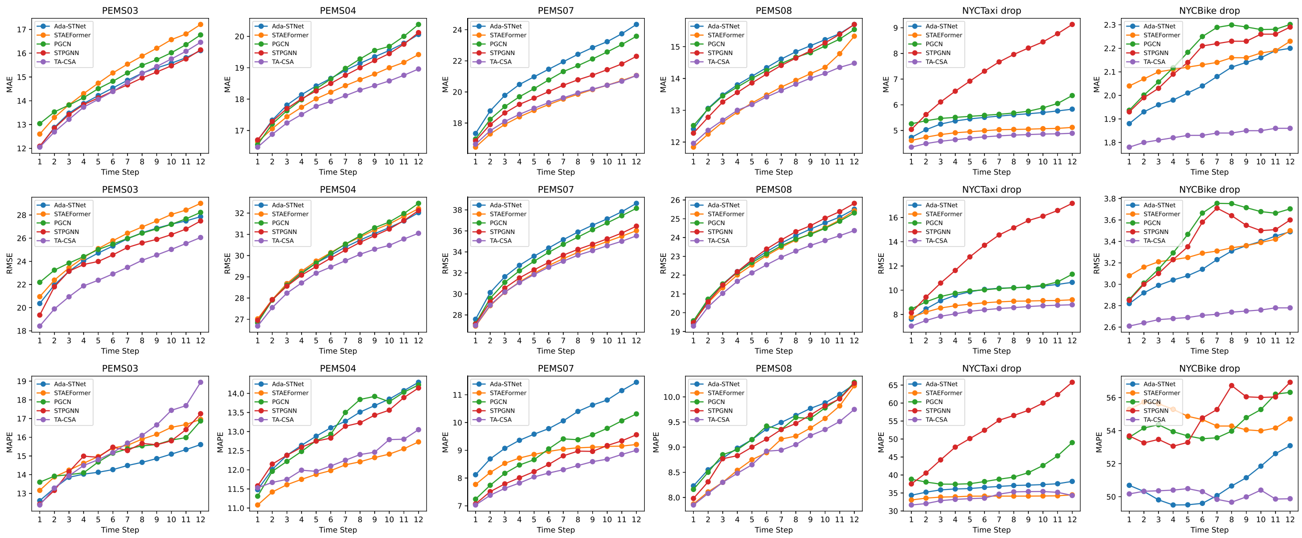

Fig. 3. Performance changes of different models on six datasets as the prediction steps increase (12 time steps).

TABLE IV

PERFORMANCE COMPARISON ON LONG-TERM TRAFFIC FORECASTING (36, 48, 60 TIME STEPS). BOLD: BEST AND UNDERLINE: SECOND BEST

Dataset	Method	@Horizon 36			@Horizon 48			@Horizon 60			Average		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
PEMS04	STGCN	23.90	37.28	16.98%	24.70	38.53	17.85%	26.43	40.80	19.45%	23.16	36.11	16.93%
	DCRNN	23.67	37.62	16.03%	24.47	38.86	16.81%	25.39	39.92	18.10%	22.79	36.19	15.70%
	ASTGCN	31.01	46.77	20.91%	33.15	49.82	23.16%	38.53	57.06	28.91%	29.97	45.76	20.54%
	AGCRN	23.33	38.34	15.48%	23.90	39.18	16.02%	25.72	41.42	17.66%	22.76	37.23	15.21%
	STG-NCDE	24.27	38.50	16.65%	24.81	39.26	17.32%	26.24	41.25	19.18%	23.56	37.23	16.45%
	ASTGNN	24.12	40.20	15.45%	24.90	41.64	16.17%	25.96	43.21	17.12%	23.00	38.34	14.90%
	STAMT	21.42	34.96	15.79%	21.89	35.67	16.05%	22.78	36.91	16.83%	20.78	33.76	15.33%
	DGCRN	23.74	37.36	16.64%	28.62	42.85	21.60%	38.68	57.36	31.15%	24.59	38.90	17.68%
	PDFormer	22.18	35.57	15.26%	22.45	36.40	15.33%	23.64	37.73	15.91%	21.34	34.55	14.60%
	STAEFormer	23.31	37.80	14.65%	23.89	38.71	15.02%	24.47	39.47	15.58%	22.41	36.36	14.29%
	PGCN	25.71	40.18	17.48%	26.51	41.10	18.28%	28.38	43.76	20.63%	24.51	38.12	17.03%
	STPGNN	25.05	38.64	17.80%	25.92	39.84	18.21%	27.58	41.99	19.78%	24.02	37.22	17.05%
	TA-CSA	**19.40**	**31.69**	**13.37%**	**20.01**	**32.57**	**14.08%**	**20.16**	**32.67**	**14.28%**	**19.86**	**32.31**	**13.91%**
PEMS08	STGCN	21.32	33.03	14.70%	23.05	35.35	15.69%	24.88	37.88	17.19%	20.47	31.64	14.05%
	DCRNN	20.48	31.73	13.73%	21.38	33.03	14.56%	23.12	35.22	16.35%	20.02	30.93	13.58%
	ASTGCN	29.32	42.91	19.76%	31.63	46.27	21.19%	36.10	52.47	25.35%	27.21	40.87	18.55%
	AGCRN	21.29	33.84	14.50%	22.53	35.78	15.52%	24.73	38.79	17.15%	20.57	32.75	13.93%
	STG-NCDE	22.20	34.28	15.99%	22.73	35.32	15.93%	23.57	36.52	16.67%	21.00	32.58	14.99%
	ASTGNN	21.23	33.38	14.82%	22.22	35.09	15.87%	23.55	36.56	17.06%	20.58	32.29	14.54%
	STAMT	17.31	28.60	12.12%	17.94	29.52	12.78%	18.60	30.43	13.49%	16.67	27.36	11.74%
	DGCRN	18.84	30.99	12.65%	23.68	36.47	16.83%	33.45	49.61	26.47%	19.67	32.35	13.62%
	PDFormer	17.61	28.98	12.22%	18.24	29.87	12.80%	19.52	31.40	14.12%	16.93	28.05	11.83%
	STAEFormer	17.62	29.39	13.09%	17.89	29.93	12.33%	18.82	30.76	13.83%	16.95	28.26	12.01%
	PGCN	19.82	31.99	13.42%	20.86	33.60	14.17%	22.30	35.59	15.66%	19.20	30.74	13.42%
	STPGNN	19.56	31.79	13.03%	20.46	33.29	13.93%	21.92	35.21	15.17%	18.98	30.70	12.80%
	TA-CSA	**15.22**	**25.35**	**10.28%**	**15.77**	**26.19**	**10.75%**	**16.12**	**26.63**	**11.17%**	**15.70**	**26.06**	**10.73%**

TABLE V

COMPARISON OF ERRORS CHANGE FROM SHORT-TERM (12 TIME STEPS) TRANSITIONING TO LONG-TERM (36, 48, 60 TIME STEPS) PREDICTIONS. BOLD: BEST

Dataset	Method	@Horizon 36			@Horizon 48			@Horizon 60		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
PEMS04	PDFormer	21.53%	19.04%	22.67%	23.01%	21.82%	23.23%	29.53%	26.27%	27.89%
	STAEFormer	28.01%	25.54%	22.08%	31.19%	28.56%	25.17%	34.38%	31.09%	29.83%
	PGCN	37.63%	33.58%	33.74%	41.92%	36.64%	39.86%	51.93%	45.48%	57.84%
	STPGNN	34.89%	29.27%	37.24%	39.58%	33.29%	40.40%	48.52%	40.48%	52.51%
	TA-CSA	**8.32%**	**7.97%**	**9.32%**	**11.73%**	**10.97%**	**15.13%**	**12.56%**	**11.31%**	**16.76%**
PEMS08	PDFormer	29.68%	23.32%	35.18%	34.32%	27.11%	41.59%	43.74%	33.62%	56.19%
	STAEFormer	30.04%	27.73%	45.77%	32.03%	30.07%	37.31%	38.89	33.68%	54.01%
	PGCN	39.38%	39.09%	44.30%	46.69%	46.09%	52.37%	56.82%	54.74%	68.39%
	STPGNN	38.04%	36.73%	41.48%	44.39%	43.18%	51.25%	54.69%	51.44%	64.71%
	TA-CSA	**13.41%**	**13.07%**	**16.29%**	**17.51%**	**16.82%**	**21.61%**	**20.12%**	**18.78%**	**26.36%**

prediction horizon, we visualize the results, as shown in Fig. 4. TA-CSA has the smallest error increment, indicating its strong stability and generalization ability. For most attention models, anomalous information at any node may spread throughout

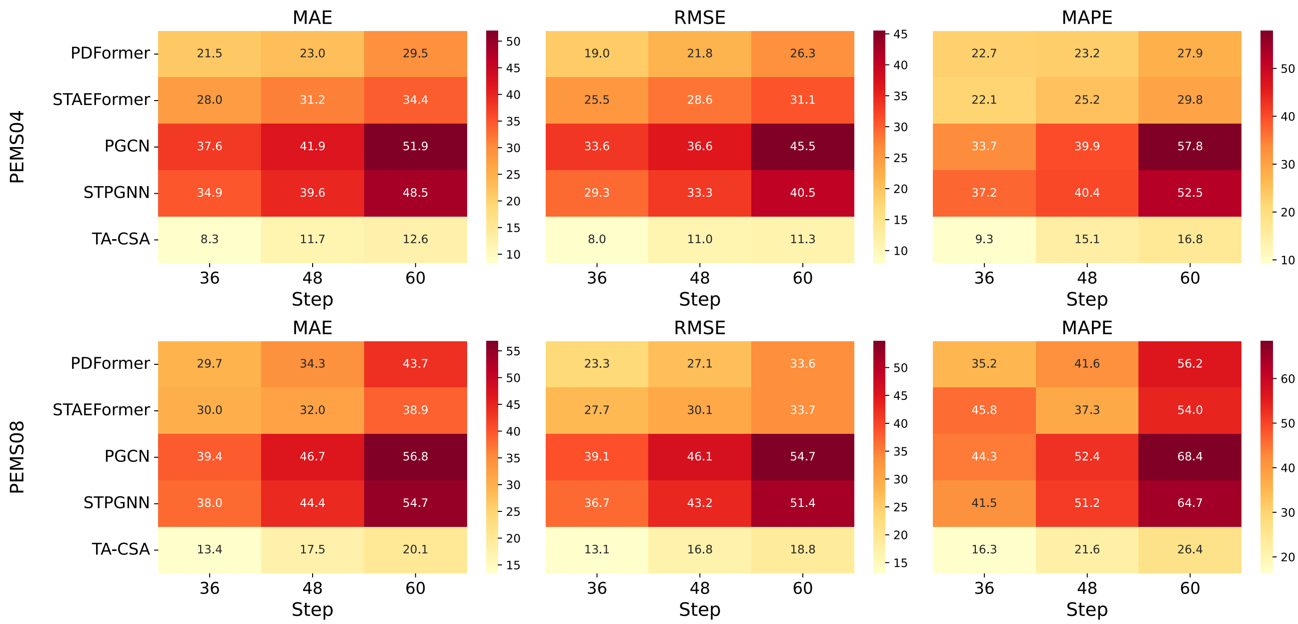

Fig. 4. Comparison of incremental errors for different models from short-term to long-term predictions. Darker colors indicate larger error increments.

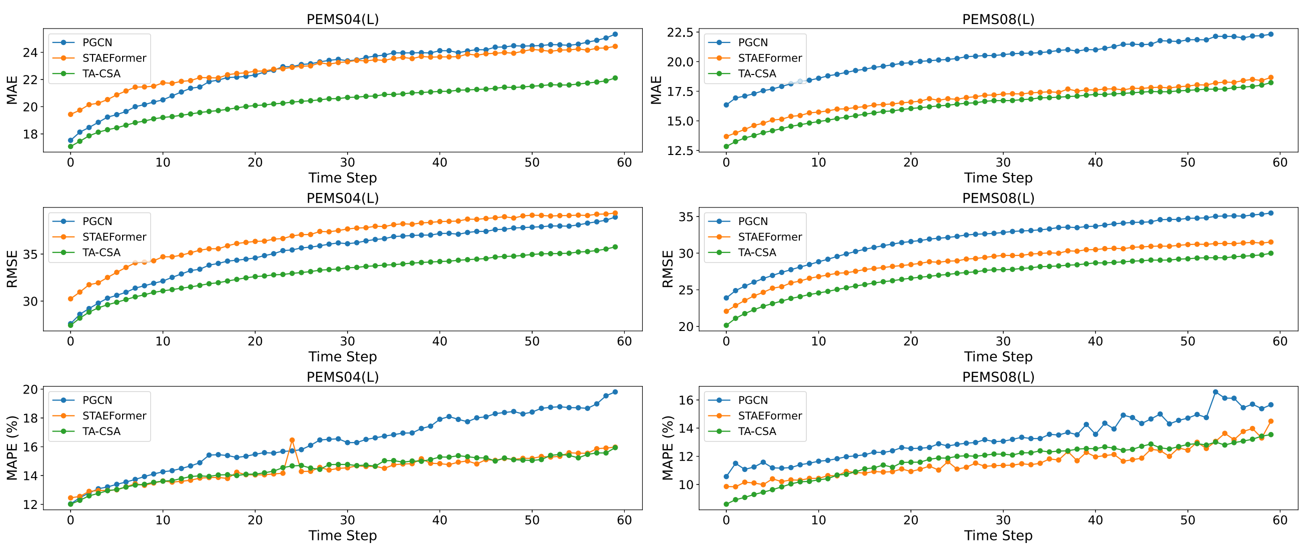

Fig. 5. Performance changes of different models on three datasets as the prediction steps increase (60 time steps).

the graph structure, enlarging the effect of errors and thus significantly weakening the prediction performance and stability of the model under anomalous conditions. The CSA module divides the nodes with functional features into multiple clusters and conducts attention modeling within each cluster, thereby restricting the spreading range and enhancing the robustness of the model. Meanwhile, the time aggregation pool module can aggregate multiscale temporal features, enabling the model to capture the characteristics of traffic data at different time granularities and enhancing its ability to model long-term dependencies. In Fig. 5, we visualize the prediction errors of the model at each time step. The results show that our model has smoother error fluctuations over the entire time series, further validating its stability and generalizability in long-term traffic prediction tasks.

F. *Efficiency Study (RQ2)*

As traffic prediction research continues to advance, the structure of prediction models has become increasingly complex. These technological advances have significantly improved the models' adaptability and prediction accuracy. However, computational efficiency and resource consumption issues are also of increasing concern, especially in long-term prediction tasks. Accordingly, how to achieve a reasonable balance between prediction accuracy and computational efficiency has become an important issue in current research. Therefore, in Fig. 6, we compare TA-CSA with six state-of-the-art traffic prediction methods on four highway datasets to evaluate their computational efficiency performance in regular and long-term prediction tasks.

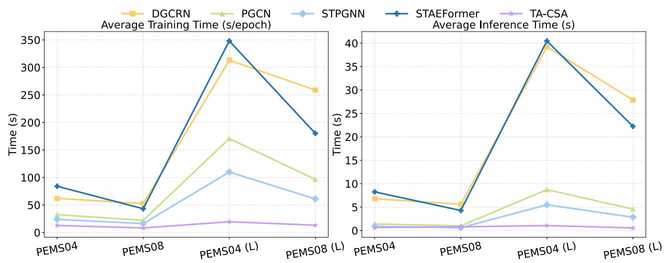

Fig. 6. Computational efficiency comparison of highway traffic datasets on the RTX 3090. PEMS04 (L) and PEMS08 (L) denote the long-term traffic forecasting tasks. Training time: Average training time per epoch. Inference time: Average time spent on each forward inference.

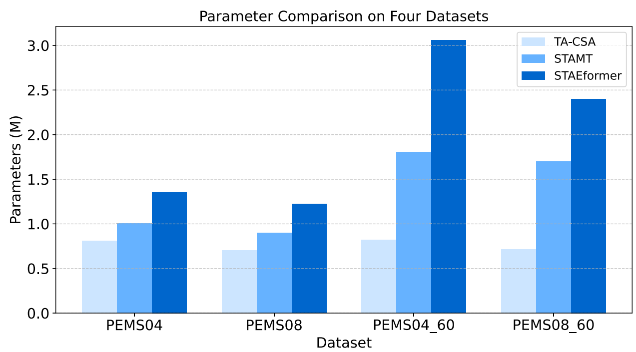

Fig. 7. Number of parameters comparison on four datasets.

The experimental results show that STG-NCDE exhibits significant training and inference overheads in both regular and long-term tasks, indicating the high computational resources required for its model structure and parameter scales. STPGNN maintains a relative balance between efficiency and performance, exhibiting moderate resource consumption and prediction effectiveness. Despite its computational efficiency, PGCN significantly lags behind STAEFormer and TA-CSA in terms of prediction accuracy. STAEFormer is more competitive in long-term prediction tasks, but this relies on higher computational costs. In terms of short-term forecasting, both STAMT and TA-CSA demonstrate competitive performance and efficiency. However, TA-CSA demonstrates the shortest training and inference times for long-term forecasting tasks, achieving the optimal balance between prediction performance and computational efficiency.

To further evaluate the model's memory resource consumption, we visualize the number of model parameters, as shown in Fig. 7. TA-CSA has a significantly smaller number of parameters. This improvement primarily results from the TAP module's ability to aggregate temporal features, thereby reducing both the number of parameters and computational load of the subsequent modules. In contrast, STAMT exhibits higher parameter complexity, primarily due to its memory bank mechanism, which requires storing historical traffic patterns and relies on numerous parameterized components for pattern management and retrieval. STAEFormer employs multilayer spatiotemporal Transformers. Each layer independently contains high-dimensional self-attention, which results in a large number of parameters and limits the computational efficiency improvements.

G. Ablation Study (RQ3)

To demonstrate the validity of the different modules in TA-CSA, we compare TA-CSA with the following variants.

1) *w/o TE:* Remove the temporal embedding layer.
2) *w/o TAP:* Remove Temporal Aggregation Pooling.
3) *w/o CSA:* Remove clustered spatial attention.
4) *w/o GLU:* Remove the gated linear unit.

We conducted ablation studies on six real traffic datasets, and the experimental results are shown in Fig. 8. The results show that the following list.

1) Removing TE leads to an increase in prediction error for all datasets. TE introduces the temporal context of "day" and "week" so that the model can perceive the periodicity in the timestamps. The lack of TE makes it difficult to model long-term time-dependent dependencies, especially in traffic flow scenarios with significant periodicity.
2) Removing TAP significantly weakens the model's performance. TAP improves the model's ability to model global trends and perceive the dynamic changes over time. Removing this module results in a weakening of the model's ability to perceive the time dimension.
3) Removing CSA will lead to a decrease in performance. CSA enables the model to divide nodes into multiple clusters, which can effectively utilize the functional similarity between nodes. Without CSA, the model would lose its ability to distinguish node heterogeneity, resulting in unstable performance in complex transportation networks.
4) Removing the GLU slightly reduces the performance, suggesting that it has an auxiliary role in optimizing the output representation.

To further validate the effectiveness and superiority of the proposed TAP and CSA modules, we conduct comparative experiments by replacing these modules with conventional methods in Fig. 9. To ensure fair comparison, all variants employ the same hyperparameter settings. The variants are as follows.

1) *TCN:* Replacing TAP with TCN, which employs a 1-D CNN to capture temporal relationships.
2) *AVG:* Remove max pooling and min pooling from the TAP, using only average pooling.
3) *MAX:* Remove average pooling and min pooling from the TAP, using only max pooling.
4) *MIN:* Remove average pooling and max pooling from the TAP, using only min pooling.
5) *GSA:* Replace CSA with global spatial self-attention to compute dependencies.
6) *GAT:* Replacing CSA with graph attention network, which uses adaptive graphs to capture potential spatial dependencies.

As shown in Fig. 9, the prediction performance of the three single pooling methods is significantly lower than TAP. This is because a single pooling method can only capture a single temporal feature. In contrast, the TAP module achieves feature fusion across the three pooling methods in the temporal dimension, which is able to capture both global and local information simultaneously. Replacing the TAP module with dilated causal convolutions results in a significant performance decline, indicating that the TAP's parallel pooling and feature

Fig. 8. Ablation study on four traffic datasets.

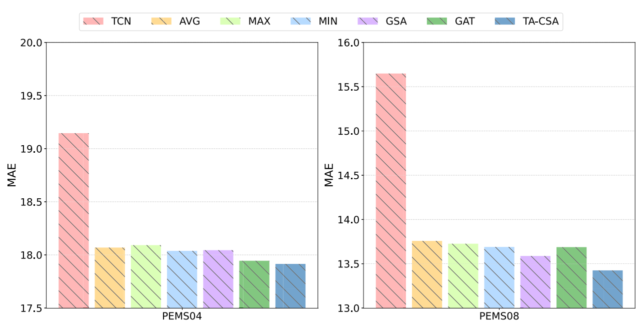

Fig. 9. Comparison of different variants.

fusion design captures temporal dependencies more effectively than single convolutional layers. To validate the effectiveness of the CSA module, we replace it with GSA and GAT, respectively. The results indicate that both perform worse than CSA. Global self-attention ignores the hierarchical structure of traffic networks, resulting in overly uniform attention. GAT relies on only fixed adjacency aggregation, making it difficult to capture dynamic spatial dependencies. In contrast, CSA effectively utilizes the spatial heterogeneity of nodes to model spatial dependencies more efficiently.

H. Effect of Pooling in TAP (RQ4)

As shown in Fig. 10, we conducted experiments on the PEMS04 and PEMS08 datasets to investigate the interpretability of global trends and dynamic events. To assess the specific contribution of these two types of information in model performance, we designed and conducted two types of comparative experiments to analyze their role and effectiveness in time-series modeling.

1) *AVG:* Removing maximum pooling and minimum pooling to explore the effect of global trends on the model.
2) *MAXMIN:* Removing average pooling to explore the effect of event dynamic fluctuations on the model.

As shown in Fig. 10(a) and (c), the experimental results show that when the TAP module only focuses on the global trend of the time series, the model can follow the trend of the real curve better in the overall trend. However, due to ignoring the transient fluctuations and dynamic changes in the event process, the model is significantly less responsive when facing critical moments, such as local peaks and sudden changes, and the prediction results show large deviations in these areas. This indicates that relying solely on average pooling to extract temporal features makes it difficult for the model to capture key dynamic features, such as local extremes, thereby limiting its ability to model local temporal patterns.

From Fig. 10(b) and (d), the experimental results show that when the TAP module focuses only on the dynamic fluctuations of events, the model can respond quickly to the information in the area of sudden changes. However, due to ignoring the global trend of the time series, it shows obvious shortcomings in overall trend modeling. This suggests that relying only on local dynamic information and lacking the guidance of the global trend will weaken the model's ability to capture the long-term evolutionary pattern of the time series, making it more susceptible to the interference of noise or short-term anomalies.

I. Parameter Study (RQ5)

To further explore the effect of the number of clusters on the performance of our proposed TA-CSA model, we design hyperparameter experiments on four publicly available traffic flow datasets. Specifically, we train and test the model and

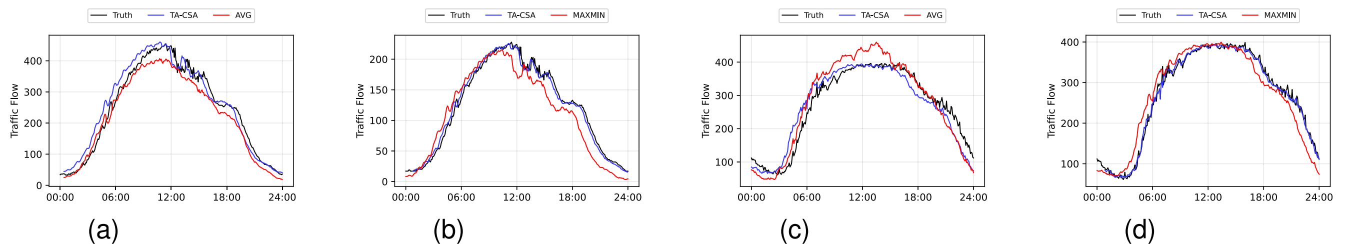

Fig. 10. Visual comparison of different inputs of TA-CSA's variants on PEMS04 and PEMS08 datasets. (a) AVGPool on PEMS04. (b) MAXMIN on PEMS04. (c) AVGPool on PEMS08. (d) MAXMIN on PEMS08.

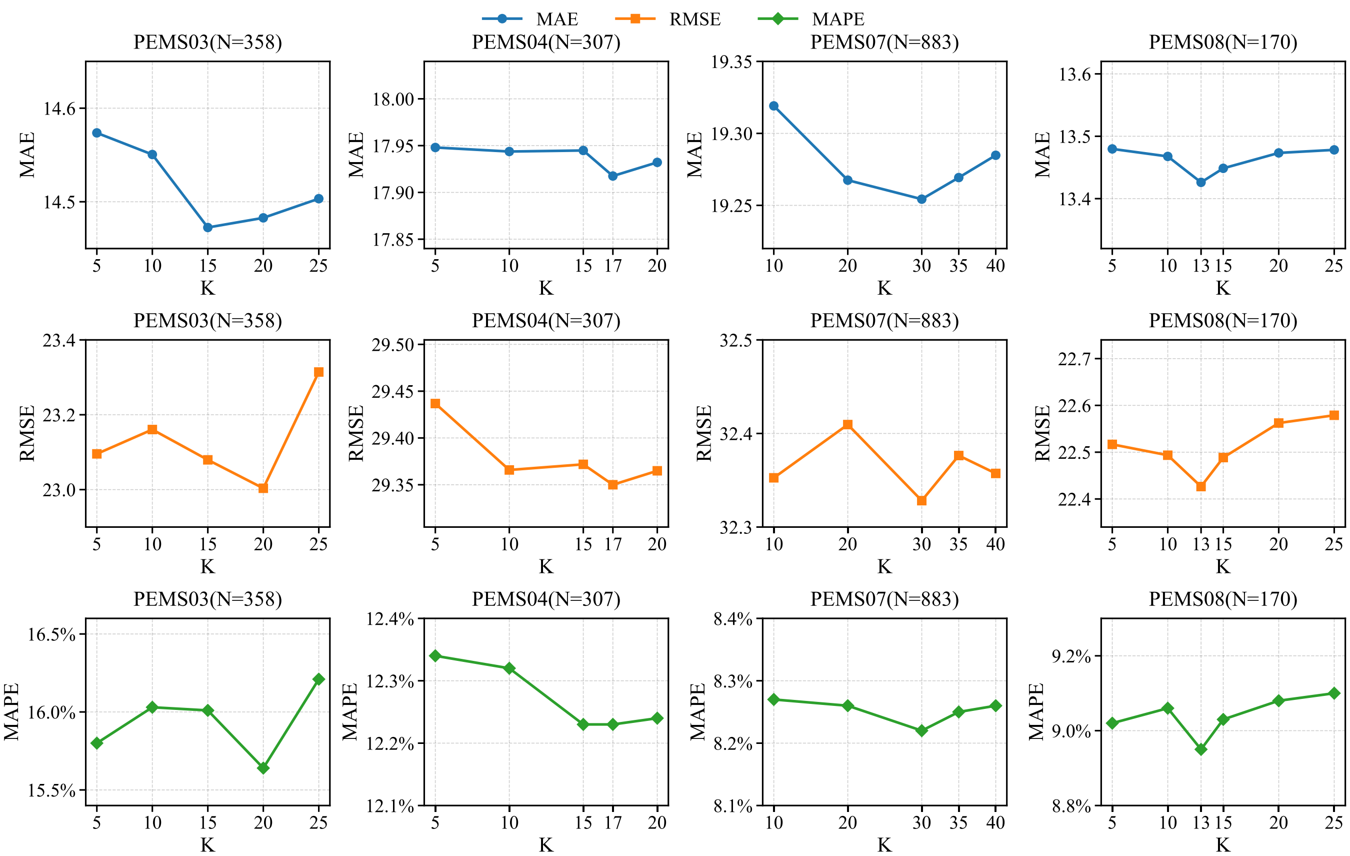

Fig. 11. Experimental results on a hyperparametric study of a traffic prediction task. Where N is the number of nodes in the dataset.

analyze how the model's performance metrics change with the number of clusters.

To analyze the relationship between hyperparameter K and node size N, we conduct experiments across multiple datasets. We test the impact of various K values on model prediction performance across datasets with differing node scales. Based on the experimental results, we plot performance variation curves across different datasets to demonstrate the model's sensitivity to K under cross-dataset conditions, as shown in Fig. 11

Experimental results indicate that the optimal K value exhibits a certain correlation with the number of nodes. When $K = (N)^{1/2}$, the model demonstrates optimal or near-optimal prediction performance across most datasets. Specifically, when the node scale N increases, raising K enables the model to utilize spatial heterogeneity among nodes better, thereby maintaining stable prediction performance. Meanwhile, setting $K = (N)^{1/2}$ effectively prevents the increase of computing costs due to excessive clusters. Conversely, when the node scale N is small, employing a smaller K value prevents noise amplification caused by overly dispersed clusters, thereby enhancing prediction performance. Based on systematic experiments across datasets of varying scales, we empirically observe that the optimal number of clusters K approximates a square root relationship with the number of nodes N ($K = (N)^{1/2}$). This relationship achieves a favorable balance between predictive performance and computational efficiency. Therefore, we adopt $(N)^{1/2}$ as the default setting for the number of clusters.

To validate the sensitivity of distance metrics, we conducted experiments using three commonly employed distance measures: Euclidean, Cosine, and Manhattan. The results are shown in Table VI. For the PEMS04 and PEMS08 datasets, the differences in MAE, RMSE, and MAPE are very small,

TABLE VI

COMPARISON OF DIFFERENT DISTANCE CALCULATION METHODS

Metric	PEMS04			PEMS08		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Euclidean	17.8752	29.3371	12.23%	13.4674	22.4492	8.99%
Cosine	17.8621	29.3316	12.17%	13.4595	22.4586	8.93%
Manhattan	17.8610	29.3280	12.16%	13.4551	22.4669	8.86%

indicating that the model is insensitive to the choice of distance metric. Therefore, we use Euclidean as the default distance metric in our experiments.

J. Rubost Testing (RQ6)

The data acquisition process is often affected by a variety of factors, such as sensor failures, communication anomalies, or environmental disturbances, which may lead to data loss or anomalies and affect data quality. Therefore, we expect the model to have strong robustness and be able to maintain stable and reliable performance in the face of these challenges. To this end, we design two robustness experiments aimed at comprehensively evaluating the model's anti-interference ability in real application environments.

1) *Data Missing:* To simulate the situation of missing data in a real environment, we remove 20%, 40%, and 60% of the data from the training set, respectively. These different missing percentages represent mild, moderate, and severe data loss scenarios to examine the impact of missing data on model performance. Meanwhile, we design two types of data loss experiments: random loss experiments model discrete data loss caused by sporadic sensor failures, while continuous loss experiments (with 12 time steps as the unit) model continuous data loss caused by prolonged sensor failures.
2) *Data Noise:* To simulate the uncertainty in real-world data, we add random Gaussian noise of different intensities to the training data, with noise proportions of 10%, 20%, and 30%, with a mean of 10 and a standard deviation of 500. In this way, we are able to assess the robustness and predictive accuracy of the model in the presence of noise disturbances.

We compare the TA-CSA model with DGCRN, PGCN, and STPGNN on the PEMS04 and PEMS08 datasets to assess the robustness of each model. The experimental results are shown in Fig. 12, and the results and analysis are as follows.

1) In the missing data experiments, TA-CSA shows excellent robustness on both PEMS04 and PEMS08 datasets. As the missing percentage increases from 0% to 60%, the MAE of TA-CSA only rises from 17.91 to 18.70 on PEMS04, with an increase of 0.79, which is significantly lower than that of DGCRN (2.00) and STPGNN (1.92). On PEMS08, the MAE of TA-CSA increased from 13.42 to 14.45 with an increase of 1.03, which was also better than DGCRN (2.95) and STPGNN (1.96). The results show that TA-CSA can effectively alleviate the performance degradation caused by missing observation data and maintain a stable prediction capability under a high missing rate environment, demonstrating good practical robustness. This is mainly attributed to the fact that TAP extracts temporal information from three aspects (global trend, local peaks, and local troughs) by fusing three operations: average pooling, maximum pooling, and minimum pooling. In the case of missing data at some time steps, TAP can still effectively mitigate the perturbation caused by missing values through information complementation, feature redundancy and dimensionality reduction integration, and enhance the model's ability to robustly model temporal patterns.
2) In the noise interference experiments, TA-CSA also shows robust noise immunity. On PEMS04, as the noise ratio increases from 0% to 30%, the MAE of TA-CSA rises from 17.91 to 19.49, with an increase of only 1.58, which is much lower than that of DGCRN (3.56) and STPGNN (1.37). On PEMS08, its MAE rises from 13.42 to 15.17 with an increase of 1.75, which is significantly better than DGCRN (5.00) and STPGNN (1.13). The TA-CSA maintains the lowest error at all noise levels, indicating its stable anti-interference capability. This is mainly attributed to the fact that the CSA module groups nodes with similar features into the same area and restricts the attention computation to the internal area. This effectively isolates the anomalous information of the nodes and prevents it from spreading to the entire graph structure, thus improving the robustness of the model in complex traffic environments.

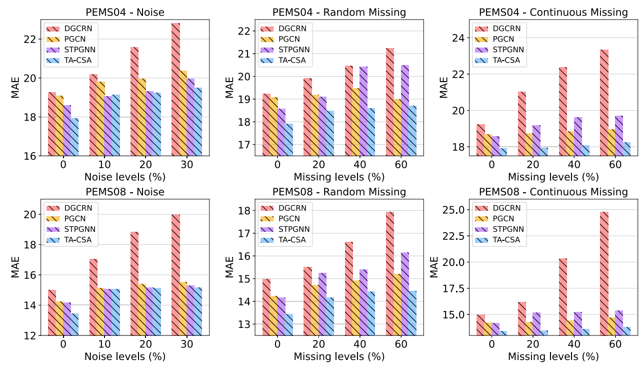

Fig. 12. Robustness testing on PEMS04 and PEMS08 datasets.

K. Significance Testing

To verify whether the performance improvement of TA-CSA over the baseline model is statistically significant, we select five random seeds (1–5) and conduct experiments on each model with the same training data and experimental conditions. The experimental results are shown in Table VII. To observe the performance distribution and stability of each, we use box plots to show the performance of different models across multiple experiments.

In Fig. 13, both the median and mean values of TA-CSA are lower than other models, while its box plot is narrower. This indicates that TA-CSA not only delivers the best performance but also exhibits higher stability under different random initializations. To assess the statistical significance of performance differences among models, we conduct *t*-tests on the results of each model. Since all models are trained and tested under identical random seeds and data splits, the experimental results are naturally paired. The *t*-tests effectively eliminate random fluctuations caused by different random seeds, thereby providing a more accurate measurement of

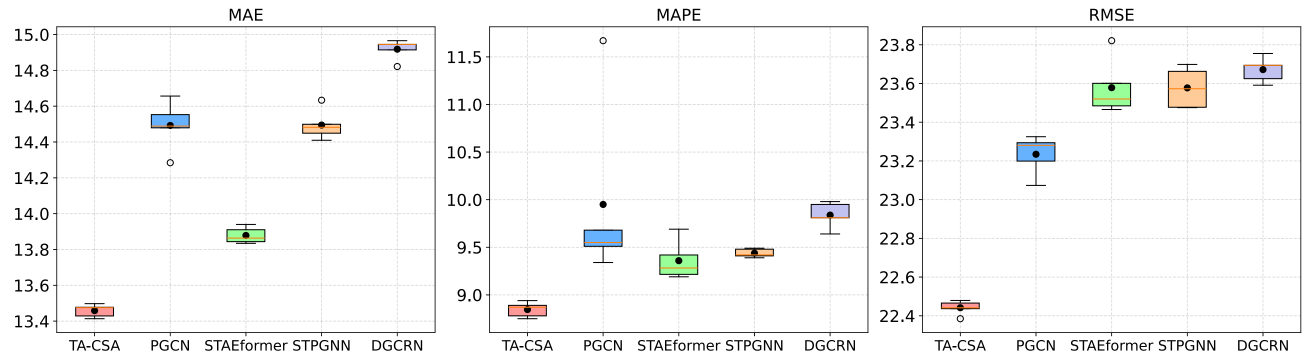

Fig. 13. Comparison of MAE distribution across models.

TABLE VII
PERFORMANCE STABILITY AND STATISTICAL SIGNIFICANCE ANALYSIS. * INDICATES STATISTICAL SIGNIFICANCE

Model	MAE(mean ± std)	95% CI	P-value
STPGNN	14.4947 ± 0.0847	/	/
DGCRN	14.9182 ± 0.0572	/	/
PGCN	14.4924 ± 0.1361	/	/
STAEformer	13.8784 ± 0.0451	/	/
TA-CSA	13.4586 ± 0.0355	/	/
TA-CSA/STPGNN	/	[0.9047, 1.1675]	**0.000013***
TA-CSA/DGCRN	/	[1.4060, 1.5133]	**0.000036***
TA-CSA/PGCN	/	[0.8347, 1.2329]	**0.000067***
TA-CSA/STAEformer	/	[0.3402, 0.4994]	**0.000063***

the performance differences between models. In addition, we calculate the 95% confidence intervals (CIs) for each model pair to quantify the reliability of the mean difference further. The CI provides a range of uncertainty for the estimation of the difference: if the CI does not contain zero, the difference can be considered statistically significant. By combining paired t-tests and CIs, we cannot only determine whether the difference is significant but also intuitively understand the magnitude of the performance difference and its CI, thereby comprehensively evaluating the actual effectiveness of the models.

The results demonstrate that TA-CSA exhibits significant performance advantages in comparison with all baseline models. Specifically, as shown in Fig. 13, the prediction error of TA-CSA is significantly lower than that of each baseline model, indicating that this model demonstrates greater performance in both prediction accuracy and stability. Meanwhile, the results for all models within the 95% CI of TA-CSA are positive and do not include zero, indicating that the performance improvement of TA-CSA is statistically significant. Furthermore, all paired t-test p-values are significantly less than 0.05, confirming that the improvements of TA-CSA over the baseline model are not due to random fluctuations but rather represent stable performance improvement. In conclusion, TA-CSA demonstrates superior performance over baseline models in terms of statistical significance, predictive accuracy, and result stability, exhibiting reliable and consistent performance.

VI. CONCLUSION

To address the challenges of maintaining a balance between performance and efficiency in long-term prediction and homogenization of spatial modeling, we propose an efficient and robust traffic prediction model, TA-CSA. Its core innovations are listed.

1) *TAP Module:* This module extracts temporal features from multiple time windows through parallel merging operations, which significantly enhances the model's ability to capture global trends and perceive dynamic changes over time.
2) *Clustered Spatial Attention:* This module utilizes hierarchical clustering to classify nodes with similar functional characteristics into multiple clusters and performs local attention computation within these clusters to achieve area-level spatial perception. Experimental validation on six real-world traffic datasets shows that TA-CSA outperforms the optimal baseline model in both regular and long-term traffic prediction tasks. We propose a new traffic prediction framework to promote research in this area, but it still has some limitations. Although the CSA module can effectively utilize the heterogeneity of nodes and improve the model's robustness, its computational complexity is $O(N^2)$, which will result in significant computational resource overhead for large-scale traffic networks. Therefore, future work will focus on designing more efficient attention mechanisms to reduce computational resource consumption while maintaining the model's prediction performance, thereby enhancing its applicability in large-scale scenarios.

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